ELITE IGCSE ACADEMY

Student Past Paper Solutions

May 2025 4WM1HR Worked Solutions

May2025 | 4WM1HR | Unit 1

29 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

1 The length of a ship is 142.8 m, correct to 1 decimal place.

(i) Write down the lower bound of the length of the ship.

.... m

(1)

(ii) Write down the upper bound of the length of the ship.

.... m

(1)

(Total for Question 1 is 2 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 2 Marks: 3

TOPIC: **Fractions**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

2 Show that $2\frac{1}{4} \times 1\frac{5}{7} = 3\frac{6}{7}$

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 3

TOPIC: **Fractions**

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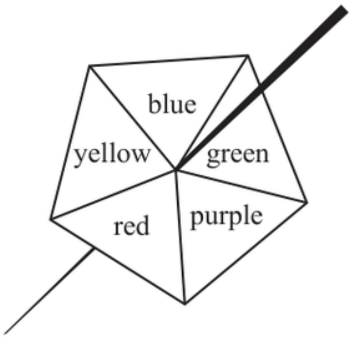
No worked solution is saved yet.

PAST PAPER QUESTION Q#: 3 Marks: 5

TOPIC: **Probability Toolkit**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

3 Here is a biased 5-sided spinner.
When the spinner is spun, it can land on blue or on green or on purple or on red or on yellow.



The table gives information about the probability of the spinner landing on each colour.

Colour	blue	green	purple	red	yellow
Probability	0.12	0.20	0.38	$4x$	x

Sophie spins the spinner once.

(a) Work out the probability that the spinner lands on blue or on green or on purple.

....

(1)

Max spins the spinner 350 times.

(b) Work out an estimate for the number of times the spinner lands on red.

....

(4)

(Total for Question 3 is 5 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 5

TOPIC: **Probability Toolkit**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 4 Marks: 4

TOPIC: **Set Notation & Venn Diagrams**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

4 $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 4, 6, 8, 10, 12\}$

$B = \{3, 6, 9, 12\}$

$C = \{1, 3, 5, 7, 9, 11\}$

(a) List the members of the set

(i) $A \cup B$

....

(ii) B'

....

(2)

$\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 4, 6, 8, 10, 12\}$

$B = \{3, 6, 9, 12\}$

$C = \{1, 3, 5, 7, 9, 11\}$

$\mathscr{E}$	$\cap$	$\cup$	$\varnothing$	$\in$	$\notin$

(b) Write a symbol from the box on each dotted line to make each of the following a true statement.

(i) $A \cap C =$ ....

(ii) 13 .... $\mathscr{E}$

(2)

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 4

TOPIC: **Set Notation & Venn Diagrams**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 5 **Marks: 4**

TOPIC: **Algebraic Roots & Indices**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

5 $3^8 \times 3^4 = 3^m$

(a) Find the value of m

$m =$ ....

(1)

$4^{12} \div 4^7 = 4^n$

(b) Find the value of n

$n =$ ....

(1)

$\left(6^2\right)^{10} = 6^p$

(c) Find the value of p

$p =$ ....

(1)

$5^r = 1$

(d) Write down the value of r

$r =$ ....

(1)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Algebraic Roots & Indices**

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PAST PAPER QUESTION

Q#: 6 Marks: 3

TOPIC: **Forming & Solving Equations**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

$\left(6^2\right)^{10} = 6^p$

(c) Find the value of p

$p =$ ....

(1)

$5^r = 1$

(d) Write down the value of r

$r =$ ....

(1)

(Total for Question 5 is 4 marks)

6 The diagram shows triangle ABC

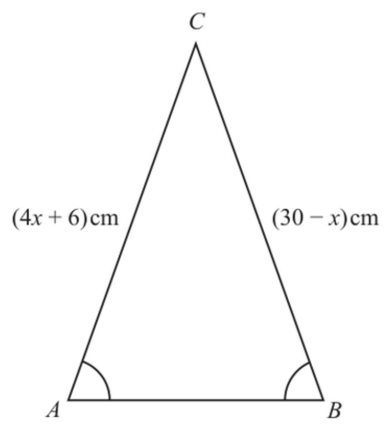

Diagram **NOT** accurately drawn

$CA = (4x + 6)$ cm $\quad CB = (30 - x)$ cm $\quad$ angle CAB = angle CBA

Work out the value of x

$x =$ ....

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 3

TOPIC: **Forming & Solving Equations**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 7 Marks: 3

TOPIC: **Standard & Compound Units**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

7 Change a speed of 40 metres per second to a speed in kilometres per hour.
Show your working clearly.

.... kilometres per hour

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Standard & Compound Units**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

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No worked solution is saved yet.

PAST PAPER QUESTION Q#: 8 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

8

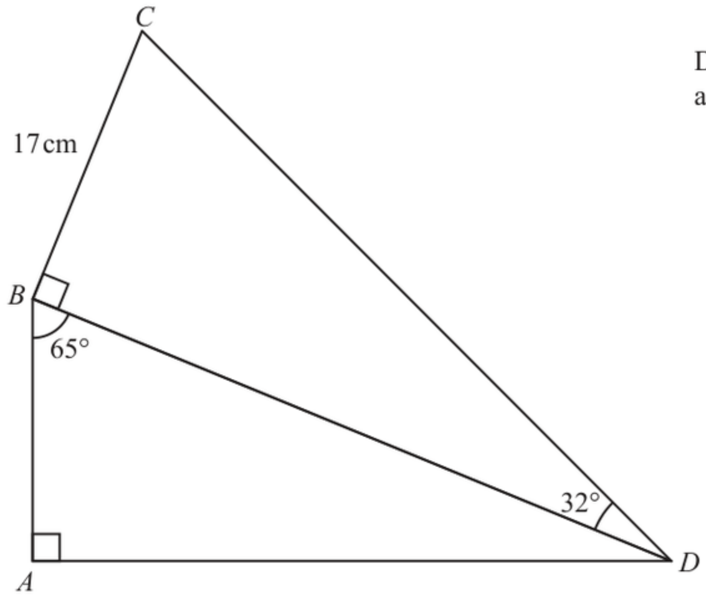

Diagram **NOT** accurately drawn

ABD and *BCD* are right-angled triangles.

angle $ABD = 65°$ angle $CDB = 32°$ $BC = 17$ cm

Work out the length of *AD*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 8 is 4 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 4

Topic: **Right-Angled Triangles - Pythagoras & Trigonometry**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 9 Marks: 7

TOPIC: **Factorising**

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9 (a) Simplify $(5k^3n^4)^2$

....
(2)

(b) Factorise fully $12m^2y^4 + 18m^3y$

....
(2)

(c) (i) Factorise $x^2 - 13x + 42$

....
(2)

(ii) Hence, solve $x^2 - 13x + 42 = 0$

....
(1)

(Total for Question 9 is 7 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 7

TOPIC: **Factorising**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 10 Marks: 5

TOPIC: **Standard & Compound Units**

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10 The diagram shows a rectangular sheet of metal.

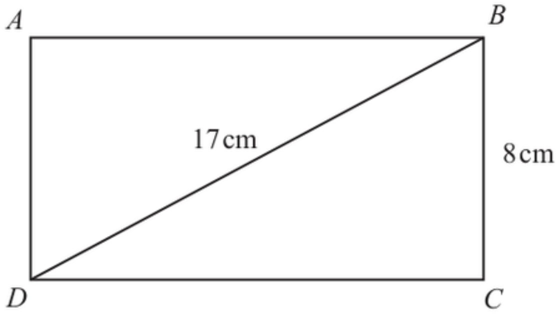

Diagram **NOT** accurately drawn

$BC = 8$ cm $\quad DB = 17$ cm

The sheet of metal lies flat on a table.
The force exerted by the sheet of metal on the table is 150 newtons.

Work out the pressure on the table due to the sheet of metal.
Show your working clearly.

$$\boxed{\text{pressure} = \frac{\text{force}}{\text{area}}}$$

.... newtons/cm^2

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 5

TOPIC: **Standard & Compound Units**

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No worked solution is saved yet.

PAST PAPER QUESTION **Q#: 11** **Marks: 4**

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

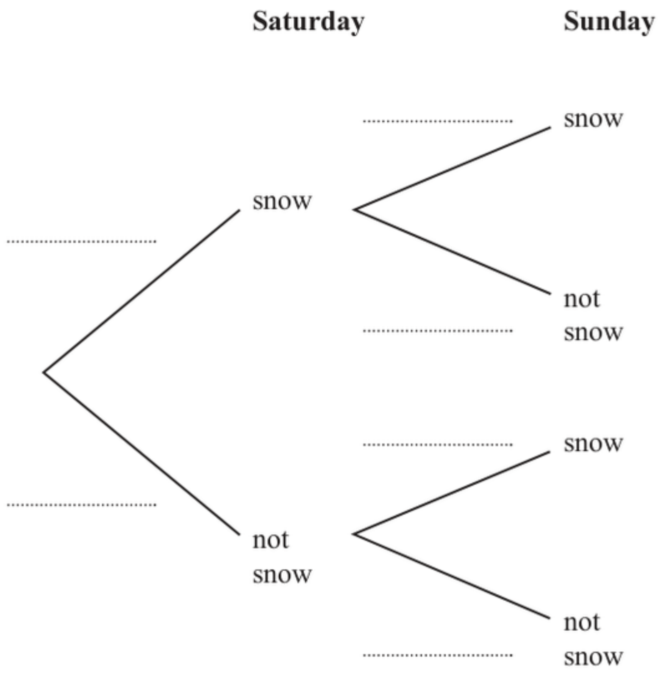
May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

11 The probability that it will snow on Saturday is 0.2

If it snows on Saturday, the probability that it will snow on Sunday is 0.35

If it does not snow on Saturday, the probability that it will snow on Sunday is 0.4

(a) Use this information to complete the probability tree diagram.



(2)

(b) Work out the probability that it will snow on both Saturday and Sunday.

....

(2)

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 12 Marks: 6

TOPIC: **Solving Linear Equations**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

12 (a) Solve $\dfrac{5a+8}{3} - \dfrac{2a+5}{4} = 23$

Show clear algebraic working.

$a =$ ....

(4)

(b) Express $\left(\dfrac{\sqrt{y}}{3}\right)^{-1}$ in the form cy^n where c and n are numbers to be found.

....

(2)

(Total for Question 12 is 6 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 6

TOPIC: **Solving Linear Equations**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

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PAST PAPER QUESTION Q#: 13 Marks: 5

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

13 In the diagram, AB represents a vertical tower and CD represents a vertical building.

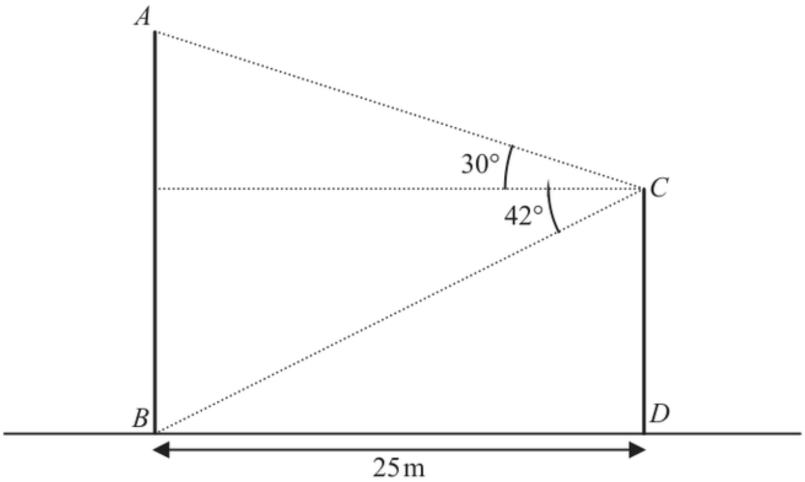

Diagram **NOT** accurately drawn

The bottom of the tower, B, and the bottom of the building, D, are on horizontal ground such that $BD = 25$ m

The angle of elevation of A from C is $30°$
The angle of depression of B from C is $42°$

(a) Work out the height of the tower AB
Give your answer correct to one decimal place.

.... m
(3)

(b) Work out the angle of elevation of A from D
Give your answer correct to one decimal place.

.... °
(2)

(Total for Question 13 is 5 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 5

Topic: **Right-Angled Triangles - Pythagoras & Trigonometry**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 14 **Marks: 2**

TOPIC: **Algebraic Proof**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

14 Use algebra to show that the recurring decimal $0.6\dot{1}\dot{2} = \dfrac{101}{165}$

(Total for Question 14 is 2 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 2

TOPIC: **Algebraic Proof**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 15 Marks: 6

TOPIC: **Expanding Brackets**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

15 (a) Solve $3x^2 + 4x - 18 = 0$
Show your working clearly.
Give your solutions correct to 3 significant figures.

....

(3)

(b) Expand and simplify $(2x-3)(x+4)(x-1)$

....

(3)

(Total for Question 15 is 6 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 6

TOPIC: **Expanding Brackets**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 16 Marks: 4

Topic: **Surds**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

16 $\sqrt{96} = 2\sqrt{k}$

(a) Find the value of k

$k =$ ....

(1)

(b) Show that $\dfrac{8}{3-\sqrt{5}}$ can be written in the form $a + \sqrt{b}$ where a and b are integers.

Show each stage of your working.

(3)

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 4

TOPIC: **Surds**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 17 Marks: 3

TOPIC: **Probability Toolkit**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

17 Kannika has 9 counters.
There is a number on each counter.

1 1 2 3 3 3 5 6 6

Kannika puts the 9 counters in a bag.
She takes at random a counter from the bag and does not replace the counter.
She then takes at random a second counter from the bag.

Work out the probability that the sum of the numbers on the two counters is less than 5

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Probability Toolkit**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 18 Marks: 4

TOPIC: **Completing the Square**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

18 (a) Express $2x^2 - 12x + 23$ in the form $a(x-b)^2 + c$ where a, b and c are integers.
Show clear algebraic working.

....

(3)

(b) Hence, state the least value of $2x^2 - 12x + 23$

....

(1)

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 4

TOPIC: **Completing the Square**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 19 **Marks: 5**

TOPIC: **Area & Perimeter**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

19 The diagram shows a pentagon $ABCDE$

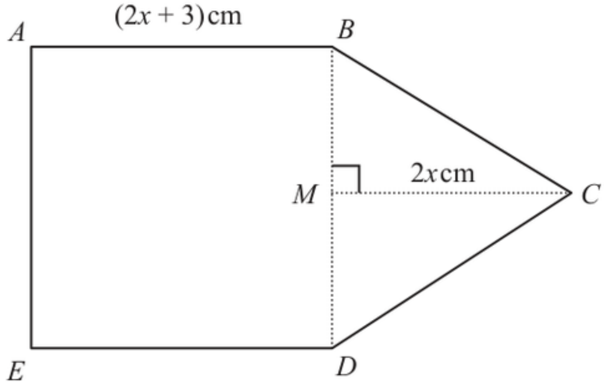

Diagram **NOT** accurately drawn

$ABCDE$ is made from a square $ABDE$ and a triangle BCD
M is the midpoint of BD

$$AB = (2x + 3)\text{ cm} \qquad MC = 2x\text{ cm} \qquad \text{angle } BMC = 90°$$

The area of pentagon $ABCDE$ is 84 cm^2

Work out the area of square $ABDE$
Show clear algebraic working.

.... cm^2

(Total for Question 19 is 5 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 5

TOPIC: **Area & Perimeter**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

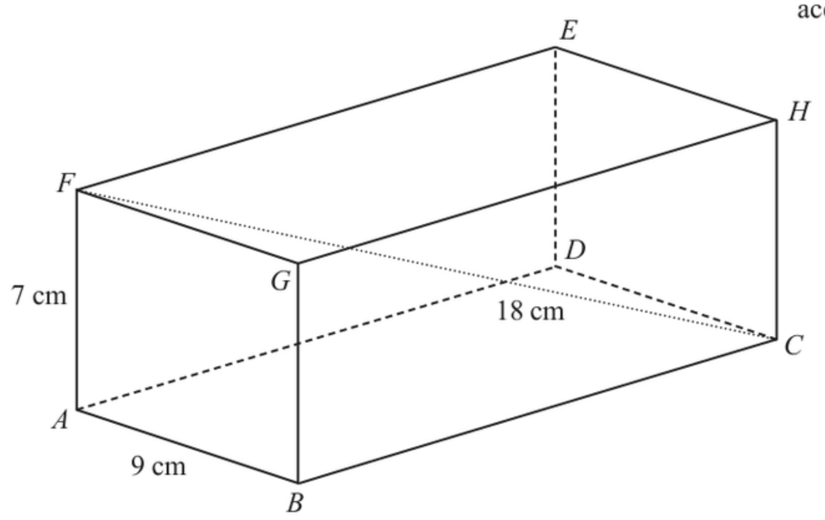
Q#: 20 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

20 The diagram shows cuboid *ABCDEFGH*

Diagram **NOT** accurately drawn



$AB = 9\,\text{cm}$ $\quad AF = 7\,\text{cm}$ $\quad FC = 18\,\text{cm}$

Calculate the length of BC

Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

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PAST PAPER QUESTION

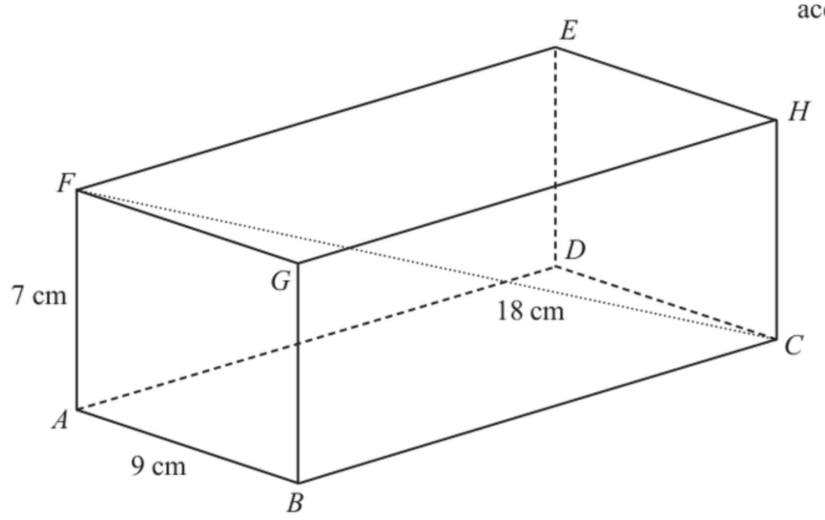
Q#: 20 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

20 The diagram shows cuboid *ABCDEFGH*

Diagram **NOT** accurately drawn



$AB = 9\,\text{cm}$ $\quad AF = 7\,\text{cm}$ $\quad FC = 18\,\text{cm}$

Calculate the length of *BC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

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No worked solution is saved yet.

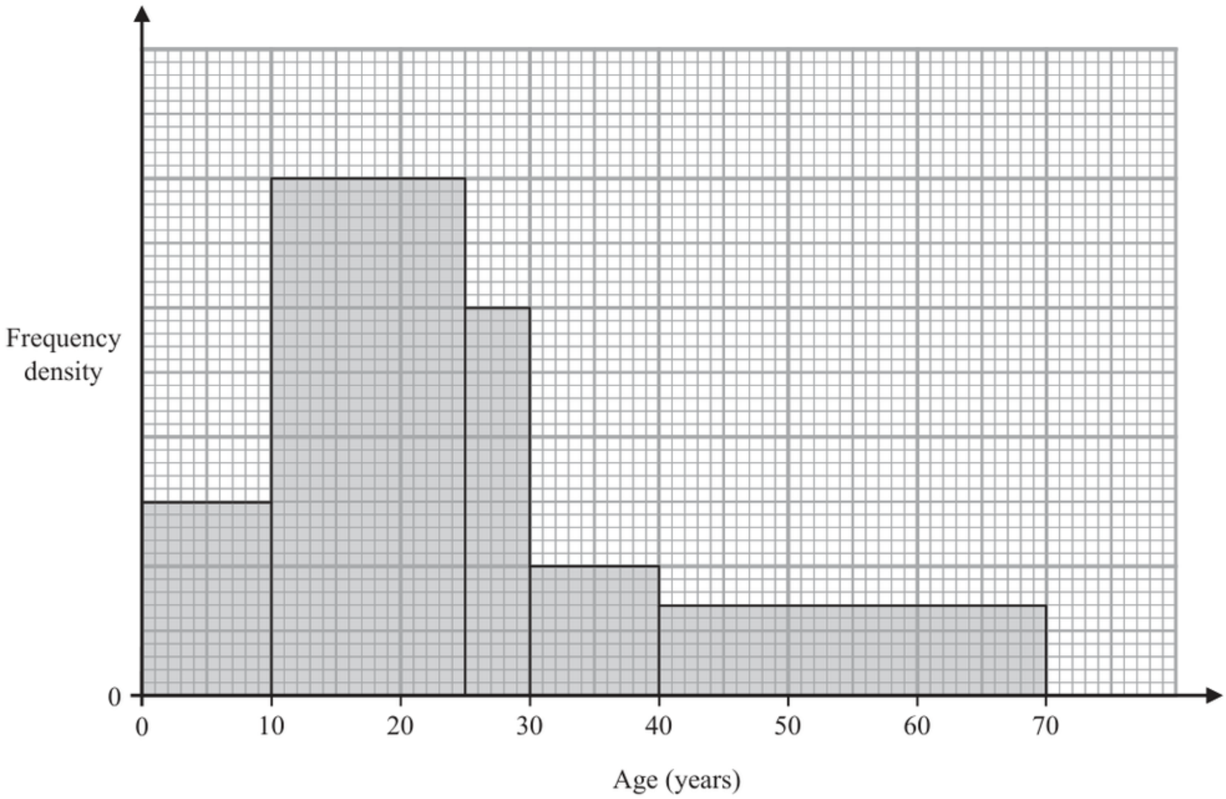
PAST PAPER QUESTION

Q#: 21 **Marks: 3**

TOPIC: **Histograms**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

21 The histogram shows information about the ages of the members of a chess club.



There are 60 members aged between 10 and 25
There are no members older than 70

Work out an estimate for the number of members who are over 35 years of age.

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Histograms**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

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No worked solution is saved yet.

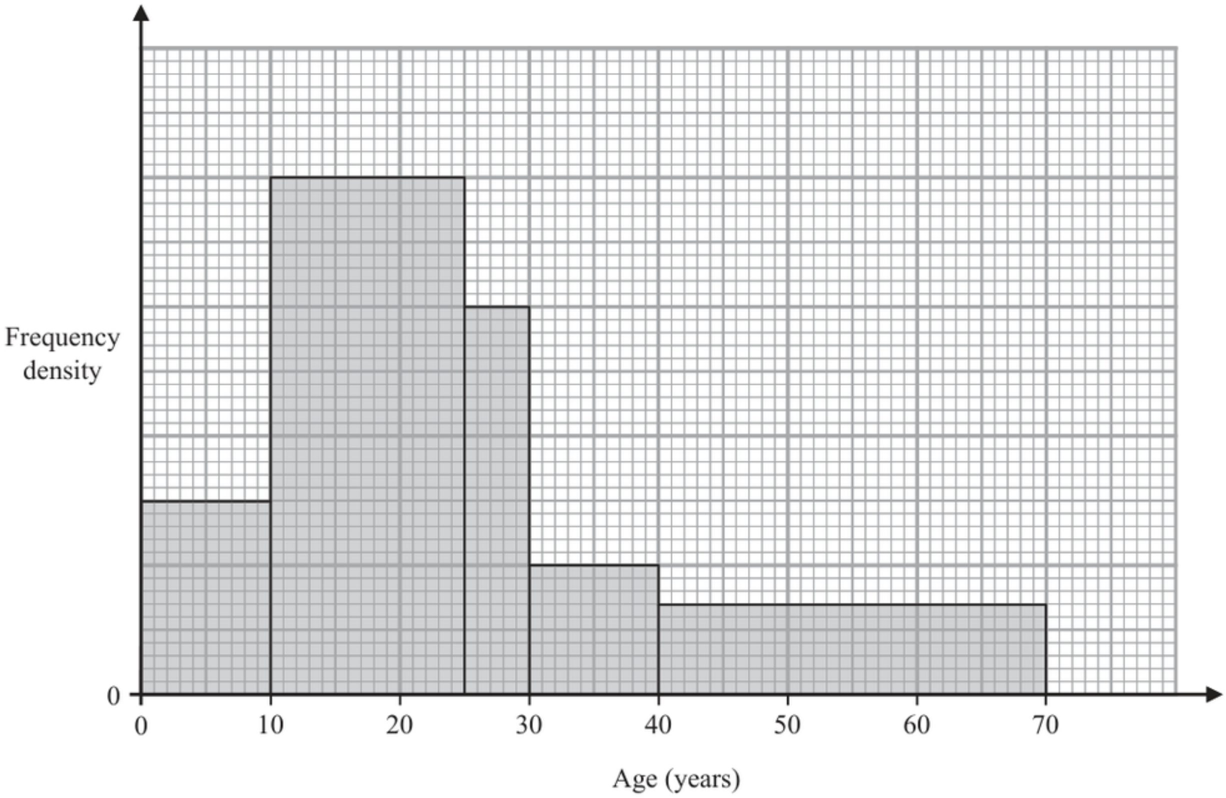
PAST PAPER QUESTION

Q#: 21 Marks: 3

Topic: **Histograms**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

21 The histogram shows information about the ages of the members of a chess club.



There are 60 members aged between 10 and 25
There are no members older than 70

Work out an estimate for the number of members who are over 35 years of age.

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Histograms**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 22 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

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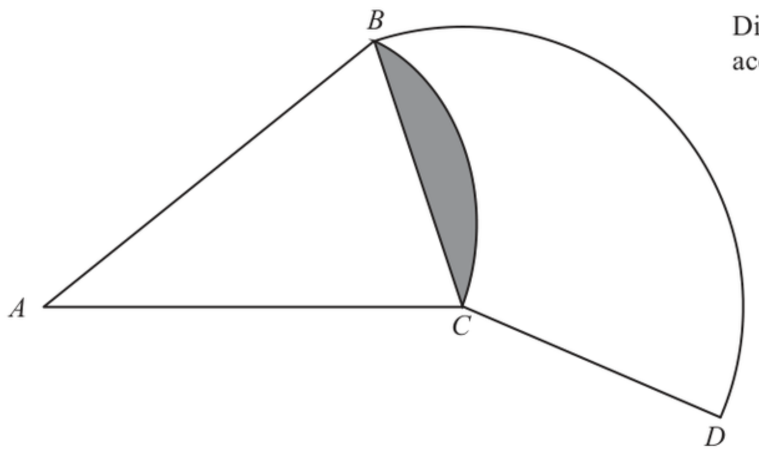
22



BAC is a sector of a circle, centre *A*

BCD is a sector of a circle, centre *C*

Angle $BAC = 40°$
Angle $BCD = 130°$
Area of shaded segment $= 28\,\text{cm}^2$

Find the length of the arc *BD*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 6

Topic: **Circles, Arcs & Sectors**

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WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 22 **Marks: 6**

TOPIC: **Circles, Arcs & Sectors**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

22

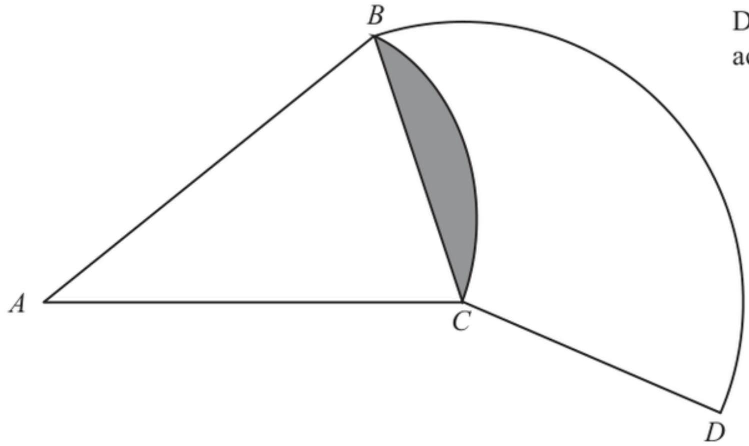

Diagram **NOT** accurately drawn

BAC is a sector of a circle, centre *A*

BCD is a sector of a circle, centre *C*

Angle $BAC = 40°$
Angle $BCD = 130°$
Area of shaded segment $= 28\,\text{cm}^2$

Find the length of the arc *BD*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 23 Marks: 5

TOPIC: **Coordinate Geometry**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

23 A circle C has centre $(3q, 4q)$ where q is an integer.

The straight line with equation $y = -\frac{5}{4}x + \frac{21}{4}$ is the tangent to C at the point $(p, p+3)$ where p is an integer.

Work out the radius of C
Give your answer in the form $\sqrt{a}$ where a is an integer.
Show your working clearly.

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Coordinate Geometry**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

WORKED SOLUTION

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 23 Marks: 5

TOPIC: **Coordinate Geometry**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

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The straight line with equation $y = -\frac{5}{4}x + \frac{21}{4}$ is the tangent to C at the point $(p, p+3)$ where p is an integer.

Work out the radius of C
Give your answer in the form $\sqrt{a}$ where a is an integer.
Show your working clearly.

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Coordinate Geometry**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 24 Marks: 4

TOPIC: **Algebraic Fractions**

May 2025 4WM1HR - May2025 | 4WM1HR | Unit 1

24 Write $(2x-1)+\frac{2x^2-5x-12}{x^2-16}\times\frac{x^2+4x}{6x^2+11x+3}$ as a single fraction in its simplest form.

Show clear algebraic working.

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 4

TOPIC: **Algebraic Fractions**

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No worked solution is saved yet.

PAST PAPER QUESTION

Q#: 24 Marks: 4

TOPIC: **Algebraic Fractions**

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24 Write $(2x-1)+\dfrac{2x^2-5x-12}{x^2-16}\times\dfrac{x^2+4x}{6x^2+11x+3}$ as a single fraction in its simplest form.

Show clear algebraic working.

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 4

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WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.